

JVM heap

spark.executor.memory = 8 GB

Reserved 300 MB

fixed • never usable by a query

Unified memory ~4.62 GB

$0.6 \times (\text{heap} - 300 \text{ MB})$ • spills to disk

Execution

~2.31 GB

join • sort • agg • shuffle

Storage

~2.31 GB

cache • broadcast

User memory ~3.08 GB

$0.4 \times (\text{heap} - 300 \text{ MB})$ • UDF / RDD objects

not tracked • OutOfMemoryError, no spill

Off-heap pool

spark.memory
.offHeap.size

second unified pool,
outside the heap,
invisible to GC

0 unless enabled

Overhead

spark.executor
.memoryOverhead
= 2 GB

thread stacks

Metaspace

netty & codec buffers

PySpark workers

actually consumed,
not a safety margin

container limit = heap + off-heap pool + overhead = 10 GB (enforced by the cluster manager; exceed it and the kernel kills the pod)